

MH2100 Calculus III

Midterm Exam 1 (Solutions)

AY 2024/2025, Semester 1

Quantitative Research Society @NTU

September 19, 2024

Overview

- **Coverage:** curves and tangent vectors; limits and differentiability; linear approximation; the Multivariable Chain Rule; tangent planes; implicit differentiation; and extrema.
- **Historical source:** MH2100 Midterm Exam 1, AY2024/2025 Semester 1. Every problem below is transcribed from the official paper. Solutions and marking schemes are unofficial QRS@NTU material.
- **Current administration (AY2026/2027):** Tuesday 6 October 2026, 16:30–18:20 (110 minutes), venue TBA. One double-sided typed A4 sheet is permitted; attachments and sticky notes are not.

Instructions to candidates (historical, verbatim)

- (1) DO NOT TURN OVER THE EXAM PAPER UNTIL INSTRUCTED TO DO SO.
- (2) This midterm exam consists of ONLY THREE problems, each consisting of multiple parts.
- (3) Candidates are required to attempt ALL THREE problems.
- (4) The total number of points for each problem is indicated at the top right-hand corner of its page.
- (5) Candidates may write anywhere on the blank pages reserved for showing work.
- (6) A SINGLE double-sided A4-sized help-sheet with handwritten notes is permitted.

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Question 1**(15 Points)****Problem 1**

- (a) A curve C in $\mathbb{R}^2$ is parametrized by the vector function $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^2$, defined by

$$\mathbf{r}(t) = (t^2 - 1, t^3 - t).$$

[4 points] Find all the unit tangent vectors to C produced by $\mathbf{r}$ at the point $(0, 0)$ on C .

- (b) For each of the following limits, either evaluate it or explain why it does not exist.

- (i) [2 points]

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2}.$$

- (ii) [2 points]

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^3}{x^2 + y^6}.$$

- (iii) [2 points]

$$\lim_{(x,y) \rightarrow (1,2)} \frac{\ln(x)}{4x^2 - y^2}.$$

- (c) Define a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = x^2 + \sin(xy).$$

[5 points] Find all the unit vectors $\mathbf{u}$ in $\mathbb{R}^2$ such that $(D_{\mathbf{u}}(f))(1, 0) = 1$.

Solution**Method 1: Direct calculation**

- (a) The first component of $\mathbf{r}(t) = (0, 0)$ gives $t = \pm 1$, and both values also make the second component zero. Since

$$\mathbf{r}'(t) = \langle 2t, 3t^2 - 1 \rangle,$$

we have $\mathbf{r}'(1) = \langle 2, 2 \rangle$ and $\mathbf{r}'(-1) = \langle -2, 2 \rangle$. These parameter values produce the two oriented unit tangents

$$\frac{1}{\sqrt{2}} \langle 1, 1 \rangle \quad \text{and} \quad \frac{1}{\sqrt{2}} \langle -1, 1 \rangle.$$

- (b) (i)

$$\left| \frac{x^2 y}{x^2 + y^2} \right| \leq |y| \rightarrow 0.$$

Thus $\boxed{0}$.

- (ii) Along $x = 0$, the quotient is 0. Along $x = y^3$ with $y \neq 0$, it equals $1/2$. Hence

$\boxed{\text{the limit does not exist.}}$

- (iii) Along $x = 1$, the quotient is 0. Along $y = 2$,

$$\lim_{x \rightarrow 1} \frac{\ln x}{4x^2 - 4} = \lim_{x \rightarrow 1} \frac{1/x}{8x} = \frac{1}{8}.$$

The path limits differ, so $\boxed{\text{the limit does not exist.}}$

(c)

$$\nabla f(x, y) = \langle 2x + y \cos(xy), x \cos(xy) \rangle, \quad \nabla f(1, 0) = \langle 2, 1 \rangle.$$

For $\mathbf{u} = \langle a, b \rangle$, the requirements are $2a + b = 1$ and $a^2 + b^2 = 1$. Substitution of $b = 1 - 2a$ gives $a(5a - 4) = 0$. Therefore

$$\boxed{\mathbf{u} = \langle 0, 1 \rangle \quad \text{or} \quad \mathbf{u} = \left\langle \frac{4}{5}, -\frac{3}{5} \right\rangle.}$$

UNOFFICIAL QRS marking scheme (15 points)

- (a) **(4 points)** 1: finds both parameters; 1: differentiates; 1: evaluates both velocities; 1: gives the two oriented unit tangents.
- (b) **(6 points)**
- (i) **(2 points)** 1: valid bound; 1: limit 0.
 - (ii) **(2 points)** 1: two paths; 1: nonexistence conclusion.
 - (iii) **(2 points)** 1: two valid path limits; 1: nonexistence conclusion.
- (c) **(5 points)** 1: gradient; 1: evaluates it; 1: directional-derivative condition; 1: unit condition; 1: both vectors.

Question 2**(15 Points)****Problem 2**(a) Define a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = \begin{cases} \frac{2x^2y + y^3}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0), \\ 0, & \text{if } (x, y) = (0, 0). \end{cases}$$

(i) [3 points] Determine whether f is continuous at $(0, 0)$. Justify your answer.(ii) [2 points] Evaluate $f_x(0, 0)$ and $f_y(0, 0)$, if these quantities exist.(iii) [4 points] Determine whether f is totally differentiable at $(0, 0)$. Justify your answer.(b) Define a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = e^{2x^2+3y}.$$

[3 points] Use a linear approximation of f at $(0, 0)$ to estimate the value of $f(0.01, -0.02)$.(c) Suppose that the functions $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ and $g : \mathbb{R}^3 \rightarrow \mathbb{R}$ satisfy the following properties:

- f is totally differentiable throughout $\mathbb{R}^3$.
- For all $x, y, z \in \mathbb{R}$, we have $g(x, y, z) = f(x - y, y - z, z - x)$.

[3 points] Use the Multivariable Chain Rule to show, for all $x, y, z \in \mathbb{R}$, that

$$g_x(x, y, z) + g_y(x, y, z) + g_z(x, y, z) = 0.$$

Solution**Method 1: Direct calculation**(a) (i) For $(x, y) \neq (0, 0)$, with $r = \sqrt{x^2 + y^2}$,

$$|f(x, y)| \leq \frac{2|x|^2|y| + |y|^3}{r^2} \leq 3r \rightarrow 0.$$

Since $f(0, 0) = 0$. Therefore

$$\boxed{f \text{ is continuous at } (0, 0).}$$

(ii) From $f(h, 0) = 0$ and $f(0, h) = h$ for $h \neq 0$,

$$\boxed{f_x(0, 0) = 0, \quad f_y(0, 0) = 1.}$$

(iii) If f were totally differentiable, its derivative would be $L(x, y) = y$. Along $(x, y) = (t, t)$,

$$f(t, t) - L(t, t) = \frac{3t^3}{2t^2} - t = \frac{t}{2}.$$

Thus the remainder divided by $\sqrt{x^2 + y^2}$ does not tend to 0. Hence

$$\boxed{f \text{ is not totally differentiable at } (0, 0).}$$

- (b) At $(0, 0)$, $f = 1$, $f_x = 0$, and $f_y = 3$. The linear approximation is

$$L(x, y) = 1 + 3y.$$

Consequently

$$f(0.01, -0.02) \approx L(0.01, -0.02) = 0.94.$$

- (c) Put $A = x - y$, $B = y - z$, and $C = z - x$, and write the partials of f at (A, B, C) as f_1, f_2, f_3 . The Chain Rule gives

$$g_x = f_1 - f_3, \quad g_y = -f_1 + f_2, \quad g_z = -f_2 + f_3.$$

Adding the three equations gives

$$g_x(x, y, z) + g_y(x, y, z) + g_z(x, y, z) = 0.$$

UNOFFICIAL QRS marking scheme (15 points)

- (a) **(9 points)**
- (i) **(3 points)** 2: valid bound; 1: continuity.
 - (ii) **(2 points)** 1: each partial derivative.
 - (iii) **(4 points)** 1: candidate derivative; 2: diagonal remainder; 1: non-differentiability conclusion.
- (b) **(3 points)** 1: derivatives at the base point; 1: linear approximation; 1: estimate 0.94.
- (c) **(3 points)** 2: three Chain Rule derivatives; 1: sums them to zero.

Question 3**(20 Points)****Problem 3**

- (a) [4 points] Find a linear equation of the tangent plane to the surface
- S
- in
- $\mathbb{R}^3$
- , given by

$$xy^2z^3 + x^4y^3z^2 = 4,$$

at the point $(1, -1, 2)$.

- (b) Define a function
- $F : \mathbb{R}^3 \rightarrow \mathbb{R}$
- by

$$F(x, y, z) = x^3 + z^2 + ye^{xz} + z \cos(y).$$

Let $S = \{(x, y, z) \in \mathbb{R}^3 \mid F(x, y, z) = 0\}$, which is the level surface of F with value 0.

- (i) [2 points] Show that the quantity
- $F_z(0, 0, 0)$
- exists and is non-zero.
- Remark (safe to ignore):**
- By the Implicit Function Theorem, we can thus find

- a sufficiently small open disk D in $\mathbb{R}^2$ centered at $(0, 0)$,
- a sufficiently small bounded open interval I in $\mathbb{R}$ centered at 0, and
- a function $z : D \rightarrow I$ that is totally differentiable throughout D ,

such that, within the small region $D \times I$, the surface S is actually the graph surface of z .

- (ii) [4 points] Use the method of implicit differentiation to evaluate
- $\frac{\partial z}{\partial x}(0, 0)$
- and
- $\frac{\partial z}{\partial y}(0, 0)$
- .

- (c) Define a function
- $f : \mathbb{R}^2 \rightarrow \mathbb{R}$
- by

$$f(x, y) = 12x^2 + y^3 - 12xy.$$

- (i) [3 points] Find all the critical points of f .
- (ii) [4 points] Use the Second-Derivative Test to determine the nature of each critical point of f .
- (iii) [3 points] Determine whether f has any absolute extremum. Justify your answer.

Solution**Method 1: Direct calculation**

- (a) Let
- $G(x, y, z) = xy^2z^3 + x^4y^3z^2$
- . Then

$$G_x = y^2z^3 + 4x^3y^3z^2, \quad G_y = 2xy^2z^3 + 3x^4y^2z^2, \quad G_z = 3xy^2z^2 + 2x^4y^3z.$$

At $(1, -1, 2)$, $\nabla G = \langle -8, -4, 8 \rangle$. Hence

$$-8(x - 1) - 4(y + 1) + 8(z - 2) = 0,$$

or equivalently

$$\boxed{-2x - y + 2z = 3.}$$

- (b) (i)

$$F_z = 2z + xye^{xz} + \cos y, \quad F_z(0, 0, 0) = 1 \neq 0.$$

Thus

$$\boxed{F_z(0, 0, 0) \text{ exists and is non-zero.}}$$

(ii) The relevant derivatives are

$$F_x = 3x^2 + yze^{xz}, \quad F_y = e^{xz} - z \sin y.$$

Implicit differentiation gives $z_x = -F_x/F_z$ and $z_y = -F_y/F_z$. At $(0, 0, 0)$,

$$\boxed{\frac{\partial z}{\partial x}(0, 0) = 0, \quad \frac{\partial z}{\partial y}(0, 0) = -1.}$$

(c) (i)

$$f_x = 24x - 12y, \quad f_y = 3y^2 - 12x.$$

Solving $y = 2x$ and $y^2 = 4x$ gives

$$\boxed{(0, 0) \quad \text{and} \quad (1, 2).}$$

(ii) The Hessian data are

$$f_{xx} = 24, \quad f_{xy} = -12, \quad f_{yy} = 6y, \quad D = 144(y - 1).$$

At $(0, 0)$, $D < 0$, so it is a saddle point. At $(1, 2)$, $D > 0$ and $f_{xx} > 0$, so it is a local minimum:

$$\boxed{(0, 0) \text{ is a saddle point; } (1, 2) \text{ is a local minimum.}}$$

(iii) Along $x = 0$, $f(0, y) = y^3$, which is unbounded above and below. Therefore

$$\boxed{f \text{ has no absolute maximum and no absolute minimum.}}$$

UNOFFICIAL QRS marking scheme (20 points)

(a) **(4 points)** 2: gradient; 1: evaluates normal; 1: correct plane.

(b) **(6 points)**

(i) **(2 points)** 1: computes F_z ; 1: evaluates it as nonzero.

(ii) **(4 points)** 2: computes F_x, F_y ; 1: implicit-derivative formulae; 1: both values.

(c) **(10 points)**

(i) **(3 points)** 2: solves first-derivative equations; 1: both points.

(ii) **(4 points)** 1: second derivatives; 1: discriminant; 1: saddle; 1: local minimum.

(iii) **(3 points)** 2: unbounded path; 1: no absolute extrema.